

Extraction and isolation of alkaloids from *Fumaria agraria*, characterization of inhibitory effect on human lung cancer cells proliferation, ID : "B73"

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Abstract

Little information is reported on the antitumor effects of isoquinoline alkaloids, particularly protopine, a major component of *Fumaria agraria*, on lung cancer. The purpose of our study is to determine the cytotoxic effect of protopine from an extraction by fractionation of the aerial part of *Fumaria agraria* on two lung cancer cell lines, NCI-H23 and NCI-H460. The basic fraction containing protopine (60.7%) has cytotoxicity to the two lung cancer cell lines studied here. The cell line NCI-H460 is more sensitive after 72 h of treatment by protopine with an IC₅₀ of $08.5 \pm 0.09 \mu\text{M}$ than the cell line NCI-H23 (IC₅₀ = $14.8 \pm 0.03 \mu\text{M}$)

Introduction

Lung cancer is the leading cause of cancer-related deaths worldwide, largely due to its high metastatic potential and late diagnosis. Smoking is responsible for nearly 80–90% of cases. Despite advances in chemotherapy and surgery, effective single-drug treatments remain limited, prompting the search for safer anticancer agents from natural products. Isoquinoline alkaloids have shown promising pharmacological properties. Protopine, a major alkaloid of *Fumaria agraria*, exhibits antioxidant, anti-inflammatory, and antiproliferative activities. However, its effects on lung cancer cells have not yet been explored.

This study aims to evaluate the cytotoxic and antiproliferative effects of extracted protopine on NSCLC cell lines NCI-H23 and NCI-H460

Methodology

Lung tissue samples were collected from 60 patients diagnosed with lung adenocarcinoma who underwent surgical resection at the University Hospital of Amiens between 2005 and 2011. All patients were fully informed and provided written consent prior to participation. Clinicopathological parameters, including age, sex, tumor size, lymph node metastasis, tumor differentiation grade, vascular invasion, necrosis, and survival, were recorded.

The aerial parts of *Fumaria agraria* were dried, powdered, and extracted with ethanol using a Soxhlet apparatus. The crude extract was concentrated under reduced pressure and subsequently fractionated by liquid–liquid extraction. After suspension in distilled water, the extract was defatted with petroleum ether, followed by chloroform extraction to obtain the neutral fraction. The aqueous phase was acidified (pH 2.5) and re-extracted with chloroform to produce the acidic fraction. Finally, the remaining aqueous layer was basified (pH 8.5) with ammonia and extracted again with chloroform to yield the basic fraction, characterized by a high protopine content (60.7%), which was selected for biological assays.

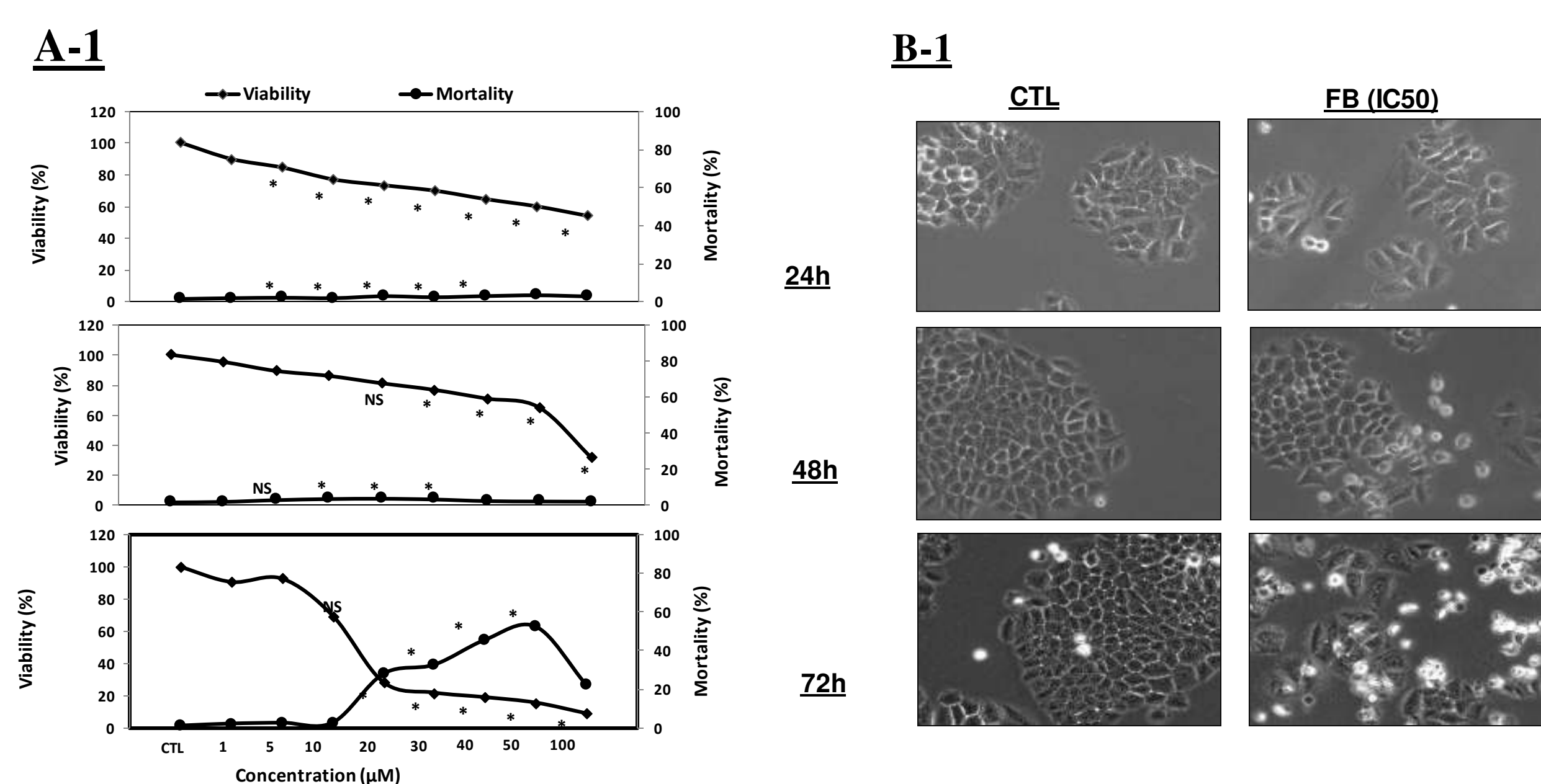
Human lung cancer cell lines NCI-H23 and NCI-H460 were cultured in Eagle's Minimum Essential Medium supplemented with 10% fetal calf serum, 2 mM L-glutamine, and 0.06% HEPES, and maintained at 37 °C in a humidified incubator containing 5% CO₂. The alkaloid-rich basic fraction was dissolved in DMSO, stored at –20 °C, and applied to cells at concentrations ranging from 1 to 100 μM .

Cell viability and mortality were evaluated using the Trypan blue exclusion assay. Cells were seeded at a density of 2×10^5 cells in 35 mm Petri dishes, incubated for 72 h, and then treated for 24, 48, and 72 h. Morphological changes were examined by phase-contrast microscopy using IC₅₀ concentrations.

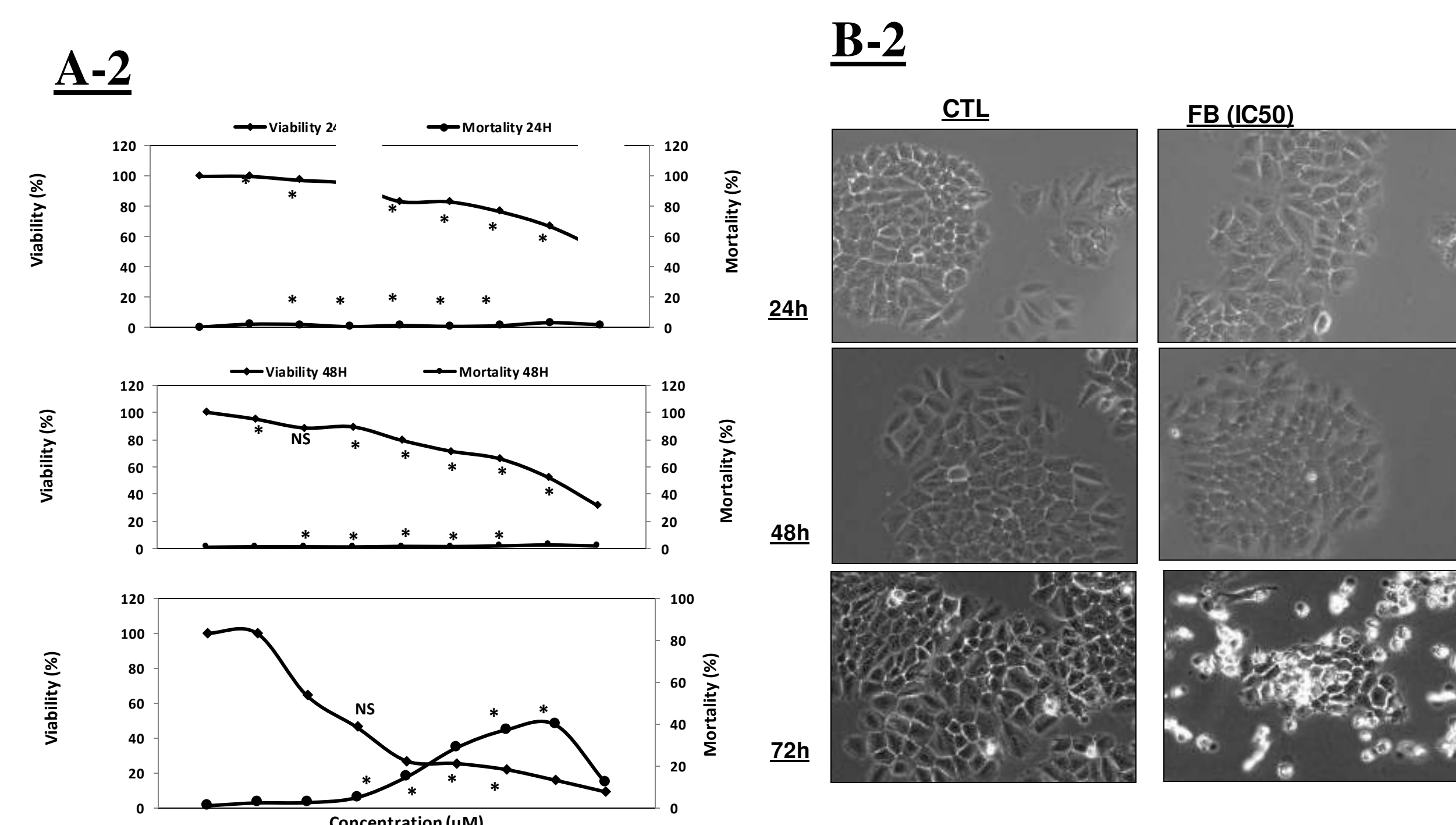
Statistical analyses were conducted using one way ANOVA or two

Results

A preliminary study on the “BF” fraction of the aerial part of *Fumaria agraria* has demonstrated its richness in protopine (60.7%) and its effectiveness on breast cancer cell lines without limited effect on normal breast cell lines [10].



Effect of the basic alkaloid fraction of the aerial part of *Fumaria agraria* “BF” on the viability, mortality and morphology of lung cancer cell lines. A. Percentage of viability and mortality of NCI-H23 cells treated with a range of FB concentrations (0, 1, 5, 10, 20, 30, 40, 50 and 100 μM) as a function of time (24 h, 48 h and 72 h). B. Photograph of the morphology of the NCI-H23 cells treated with the corresponding IC₅₀ (24 h, 48 h and 72 h), the cells were observed under a phase contrast microscope compared to the control cells treated with the same volume of DMSO (control).



Effect of the basic alkaloid fraction of the aerial part of *Fumaria agraria* “BF” on the viability, mortality and morphology of lung cancer cell lines. A. Percentage of viability and mortality of NCI-H460 cells treated with a range of FB concentrations (0, 1, 5, 10, 20, 30, 40, 50 and 100 μM) as a function of time (24 h, 48 h and 72 h). B. Photograph of the morphology of the NCI-H460 cells treated with the corresponding IC₅₀ (24 h, 48 h and 72 h), the cells were observed under a phase contrast microscope compared to the control cells treated with the same volume of DMSO (control)

Conclusion and discussion

The results obtained showed that the basic alkaloid fraction BF had an antiproliferative effect on pulmonary cancer cell lines at different degrees. The sensitivity of a cell line depends on its tissue characteristics and invasiveness.

The BF fraction, represented mainly by protopine, has proved its cytotoxic efficacy with respect to other cancerous cell lines namely breast cancer. It is important to note that the anti-cancer effect of the aerial part of *Fumaria agraria* on lung cancer cell lines has never been studied. Once again, the genus *Fumaria* proves its use as a medicinal plant.

References